

- 1 (a) Define gravitational potential at a point.

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..... [2]

- (b) A uniform spherical planet has radius  $R$ . The gravitational potential at its surface is  $-\Phi$ .

P is a point on a line that passes through the centre of the planet, at variable displacement  $x$  from the centre, as shown in Figure 1.1.

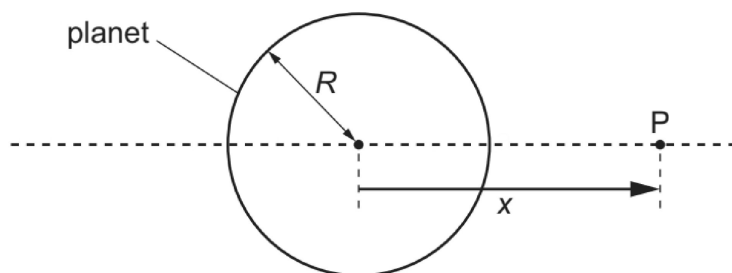


Figure 1.1

Figure 1.2 shows the variation with  $x$  of the gravitational potential  $\phi$  due to the planet in the region outside the planet close to its surface.

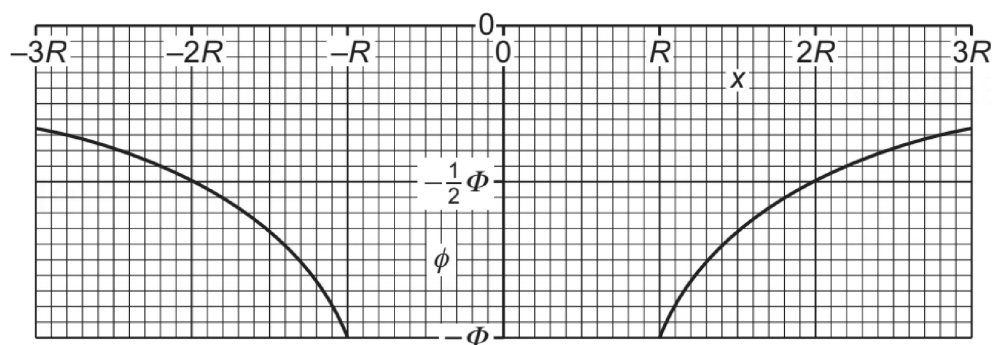


Figure 1.2

- (i) Explain why  $\phi$  is negative for all values of  $x$ .

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..... [2]

- (ii) State an expression, in terms of  $\Phi$  and  $R$ , for the mass  $M$  of the planet. Identify any other symbols that you use.

$$M = \dots\dots\dots [2]$$

- (iii) Determine an expression, in terms of  $\Phi$  and  $R$ , for the gravitational field strength  $g_0$  at the surface of the planet.

$$g_0 = \dots\dots\dots [1]$$

- (iv) On Figure 1.3, sketch the variation of the gravitational field  $g$  with  $x$  between  $x = -3R$  and  $x = 3R$ . Do **not** include the region inside the sphere between  $x = -R$  and  $x = R$ .

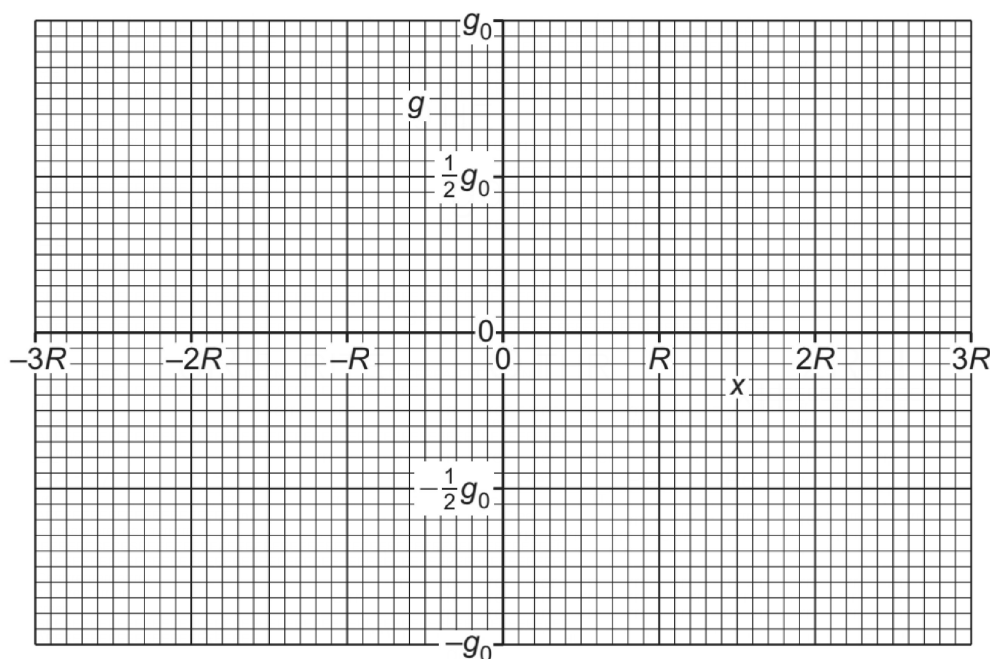


Figure 1.3

[3]

[Total: 10]